CCEF-LITE v3.0 Master Document

Ontology-Locked, Mechanism-Driven, Projection-Clean Cosmology

Allan Gibson
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Part I: Ontological Core

1 Continuum Ontology

The CCEF ontology begins with a single fundamental object: a constrained continuum field $n(x,t)$. This field is defined at every point of an underlying index space and satisfies the unit-norm condition

$$|n(x,t)| = 1. \tag{1}$$

No geometric, metric, or spacetime structure is assumed. The index variables x and t serve only as labels for internal evolution. They do not represent coordinates of an external manifold.

The continuum is the only ontological entity. All physical structure, including matter, inertia, forces, and large-scale behaviour, emerges from internal excitations, deformations, and topological configurations of $n(x,t)$. There are no particles, no external fields, and no independent background.

The vacuum configuration is the uniform state

$$n(x,t) = n_0, \tag{2}$$

which defines the reference orientation of the continuum.

1.1 Fundamental Constraint

The constraint $|n| = 1$ defines the configuration manifold as a unit sphere in the internal field space. All admissible variations of n must preserve this constraint. The continuum therefore evolves on a nonlinear manifold, and all dynamics must be compatible with the constraint.

The constraint is not imposed externally. It is an intrinsic property of the continuum and determines the allowed internal degrees of freedom.

1.2 Internal Evolution Principle

All evolution is internal to the continuum. There is no external law of motion, no background geometry, and no external time. The continuum redistributes its internal configuration according to its own variational structure.

Quantities such as energy, momentum, and pressure are emergent bookkeeping constructs that arise from coarse-grained behaviour. They are not fundamental.

2 Variational Structure

The continuum admits an internal action principle. The action is

$$S = \int d^4x \left[\frac{1}{2}\partial_\mu n \cdot \partial^\mu n - V(n) - \lambda(x,t)\left(|n|^2 - 1\right) \right], \tag{3}$$

where $V(n)$ is an internal potential and λ is a Lagrange multiplier enforcing the constraint.

The index μ labels internal directions of variation. No geometric interpretation is assigned to it.

2.1 Equation of Motion

Variation of the action yields the constrained equation of motion

$$\partial_\mu \partial^\mu n + \frac{\partial V}{\partial n} + \lambda n = 0, \tag{4}$$

with λ chosen to preserve $|n| = 1$ dynamically.

This equation describes internal redistribution of deformation energy within the continuum.

3 Linear Excitation Sector

Small deviations around the vacuum are written

$$n = n_0 + \pi, \tag{5}$$

with the constraint implying

$$n_0 \cdot \pi = 0. \tag{6}$$

To linear order, the excitations satisfy

$$\Box \pi = 0, \tag{7}$$

representing tangent-space wave modes of the constrained continuum.

These modes are not fields propagating on a geometric background. They are internal redistributions of the continuum configuration.

4 Higher-Order Continuum Structure

Beyond the linear regime, additional internal operators may appear in the effective description. These include quartic derivative terms, mixed contractions, and nonlinear constraint corrections. Such terms modify dispersion, stability, and response behaviour but remain entirely internal to the continuum.

4.1 Response Kernel Family

The internal response of the continuum is encoded in a kernel $K(k, a)$, which determines how coarse-grained density fluctuations source long-range response. The kernel is not a geometric object. It is a projection of the soliton sector and the coupling field.

The minimal kernel family is

$$K(k, a) = \frac{A(a)}{k^2 + m^2(a)} + \frac{B(a)}{k^2 + \Lambda^2(a)}, \tag{8}$$

with

$$m(a) = \frac{1}{\xi_R(a)}, \tag{9}$$

$$\Lambda(a) = \frac{1}{R_{\mathrm{sol}}(a)}, \tag{10}$$

$$A(a) = C_\alpha \, \alpha_0^2(a) \, \rho_0(a), \tag{11}$$

$$B(a) = C_\sigma \, \sigma_\alpha^2(a) \, \rho_0(a). \tag{12}$$

All quantities are internal CCEF fields. No geometric interpretation is assigned to k or a; they are labels for internal scale and coarse-graining level.

5 Topological Sector

The continuum supports stable topological configurations. For mappings into S^2, the relevant homotopy group is

$$\pi_3(S^2) = \mathbb{Z}. \tag{13}$$

Topological charge Q is an integer and labels stable soliton configurations.

Matter is not a separate ontology. It is the persistent topology of $n(x,t)$.

5.1 Stability of Solitons

Topological excitations obey Derrick-type stability conditions. The energy functional contains terms scaling differently under dilation:

$$E(\lambda) = \lambda E_2 + \lambda^{-1} E_4. \tag{14}$$

The minimum occurs at

$$\lambda^* = \sqrt{\frac{E_4}{E_2}}, \tag{15}$$

which defines the stable soliton size.

6 Emergent Inertia

A soliton with centre $X(t)$ behaves as an effective massive object. Its kinetic energy takes the form

$$T = \frac{1}{2} M \dot{X}^2, \tag{16}$$

where M is the soliton energy.

Inertia is emergent from internal deformation energy. It is not a fundamental property.

7 Long-Range Response

In dilute regimes, long-wavelength excitations satisfy

$$\nabla^2 \pi = 0, \tag{17}$$

with solutions

$$\pi(r) = \frac{B}{r}. \tag{18}$$

This produces effective inverse-distance response behaviour. It is not a force law and not geometry. It is a regime of continuum response.

Part II: Soliton Gas and Coarse-Grained Fields

8 Soliton Ensemble and Coarse-Graining

The microscopic degrees of freedom of the continuum include stable topological excitations of $n(x,t)$. These excitations form a discrete ensemble of solitons, each characterised by:

$$\{X_i(t),\ P_i(t),\ Q_i,\ M_i\}, \tag{19}$$

where X_i is the soliton centre, P_i its effective momentum, Q_i its topological charge, and M_i its emergent inertial mass.

A coarse-grained description is obtained by averaging over a smoothing volume ΔV using a kernel $W(x)$. The coarse-grained fields are:

$$\rho(x,t) = \frac{1}{\Delta V} \sum_i M_i \, W(x - X_i), \tag{20}$$

$$\sigma(x,t) = \frac{1}{\Delta V} \sum_i Q_i \, W(x - X_i), \tag{21}$$

$$u(x,t) = \frac{\sum_i P_i W(x - X_i)}{\sum_i M_i W(x - X_i)}. \tag{22}$$

These fields are not fundamental. They are projections of the soliton ensemble and serve as effective descriptors of the local state of the continuum.

8.1 Interpretation of Coarse-Grained Fields

- $\rho(x,t)$ is the inertial density of the soliton gas.
- $\sigma(x,t)$ is the response density, representing the coarse-grained contribution of soliton response content.
- $u(x,t)$ is the transport velocity, representing the coarse-grained motion of the soliton ensemble.

None of these fields are geometric or metric objects. They arise purely from internal averaging.

9 Inertial Density

The inertial density is defined as:

$$\rho(x,t) = \frac{1}{\Delta V} \sum_i M_i W(x - X_i). \tag{23}$$

This quantity measures the local concentration of soliton deformation energy. It is conserved under transport:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0. \tag{24}$$

This continuity equation follows from soliton trajectories and requires no geometric assumptions.

10 Response Density

The response density is defined as:

$$\sigma(x,t) = \frac{1}{\Delta V} \sum_i Q_i W(x - X_i). \tag{25}$$

If soliton mergers conserve total response content, then:

$$\partial_t \sigma + \nabla \cdot (\sigma u) = 0. \tag{26}$$

If creation or annihilation channels exist, a source term J_Q appears:

$$\partial_t \sigma + \nabla \cdot (\sigma u) = J_Q. \tag{27}$$

11 Coupling Field

The effective coupling per unit inertial content is:

$$\alpha(x,t) = \frac{\sigma(x,t)}{\rho(x,t)}. \tag{28}$$

This field determines how strongly a region participates in long-range response. If α is uniform, the continuum exhibits universal response. If α varies, the response becomes composition dependent.

11.1 Mean Coupling and Variance

Define:

$$\alpha_0(t) = \langle \alpha(x,t) \rangle, \tag{29}$$

$$\sigma_\alpha^2(t) = \langle (\alpha - \alpha_0)^2 \rangle. \tag{30}$$

The variance σ_α^2 plays a central role in the stochastic sector and the RG flow. It determines the strength of internal fluctuations and the degree of non-universality.

12 Transport Equations

The coarse-grained momentum equation is:

$$\rho\,(\partial_t u + u \cdot \nabla u) = -\nabla P - \rho \nabla \Phi + \nu \nabla^2 u + F_{\text{corr}}, \tag{31}$$

where:

- P is velocity-dispersion pressure,
- ν is effective viscosity,
- F_{corr} represents unresolved correlation stresses.

This equation is a projection of soliton interactions. It is not a geometric or force-based law.

13 Response-Generated Potential

The effective potential Φ is defined by the response kernel:

$$\Phi(x,t) = \int K(x - x'; t)\, \rho(x',t)\, d^3x'. \tag{32}$$

This potential is not a fundamental field. It is a projection of the continuum's internal response structure.

13.1 Homogeneous Kernel

If the kernel becomes translationally invariant under coarse-graining:

$$K(x,x') = K(|x - x'|), \tag{33}$$

then the potential becomes a convolution:

$$\Phi = K * \rho. \tag{34}$$

14 Inverse-Distance Response Regime

If the kernel approaches:

$$K(r) \sim \frac{1}{r}, \tag{35}$$

then:

$$\Phi(x) = \int \frac{\rho(x')}{|x - x'|} d^3x'. \tag{36}$$

Applying the Laplacian yields:

$$\nabla^2 \Phi = -4\pi C \rho. \tag{37}$$

This is an internal response regime, not a geometric Poisson equation.

15 Pressure and Equation of State

Define velocity fluctuations:

$$c_i = v_i - u_i. \tag{38}$$

The pressure tensor is:

$$P_{ij} = \rho \langle c_i c_j \rangle. \tag{39}$$

In isotropic regimes:

$$P_{ij} = P \delta_{ij}. \tag{40}$$

The effective equation of state is:

$$P = P(\rho, \sigma_\alpha^2, \xi_R), \tag{41}$$

determined entirely by internal continuum structure.

16 Response Correlation Length

The kernel decay defines a correlation length:

$$K(r) \sim \frac{e^{-r/\xi_R}}{r}. \tag{42}$$

If ξ_R is finite, the response is screened. If ξ_R diverges, the response becomes long-range.

17 Coupling Field Dynamics

Since $\alpha = \sigma/\rho$, its evolution obeys:

$$\partial_t \alpha + u \cdot \nabla \alpha = D_\alpha \nabla^2 \alpha + S_{\text{merge}} + S_{\text{nl}}, \tag{43}$$

where:

- D_α is a mixing coefficient,
- S_{merge} is merger-driven bias,
- S_{nl} represents nonlinear feedback.

18 Homogeneous Background State

In a uniform regime:

$$\rho = \rho_0(t), \qquad \sigma = \sigma_0(t), \qquad \alpha = \alpha_0(t). \tag{44}$$

The velocity field takes the form:

$$u = H(t)x, \tag{45}$$

defining a bulk expansion rate H.

18.1 Background Continuity

Substituting $u = Hx$ gives:

$$\dot{\rho}_0 + 3H\rho_0 = 0, \tag{46}$$

so that:

$$\rho_0 \propto a^{-3}. \tag{47}$$

This dilution arises from transport, not geometry.

19 Background Coupling Evolution

The mean coupling evolves as:

$$\dot{\alpha}_0 = -\Gamma_\alpha(\rho_0)\,\alpha_0 + S_0(\rho_0), \tag{48}$$

where Γ_α is a relaxation rate and S_0 is a source term.

20 Bulk Expansion Closure

The expansion rate satisfies:

$$H^2 = F(\rho_0, \alpha_0, P_R, \Sigma), \tag{49}$$

where P_R and Σ are internal response pressure and stress.

Acceleration occurs when:

$$P_R < -\frac{\rho_0}{3}. \tag{50}$$

This is an internal relaxation effect, not a geometric force.

Part III: Response Kernel and Projection Layer

21 Response Kernel Family

The response kernel $K(k, a)$ encodes how coarse-grained density fluctuations source long-range internal response within the continuum. It is not a geometric object and does not represent curvature or a force. It is a projection of the soliton sector, the coupling field, and the RG flow.

The minimal kernel family is defined as:

$$K(k, a) = \frac{A(a)}{k^2 + m^2(a)} + \frac{B(a)}{k^2 + \Lambda^2(a)}, \tag{51}$$

where the scale parameters are:

$$m(a) = \frac{1}{\xi_R(a)}, \tag{52}$$

$$\Lambda(a) = \frac{1}{R_{\text{sol}}(a)}. \tag{53}$$

The amplitudes are:

$$A(a) = C_\alpha \, \alpha_0^2(a) \, \rho_0(a), \tag{54}$$

$$B(a) = C_\sigma \, \sigma_\alpha^2(a) \, \rho_0(a). \tag{55}$$

All quantities are internal CCEF fields. The variable k labels internal scale, and a labels coarse-graining level. Neither carries geometric meaning.

21.1 Interpretation of Kernel Components

- The term $A(a)/(k^2+m^2)$ represents long-range response controlled by the correlation length ξ_R.
- The term $B(a)/(k^2 + \Lambda^2)$ represents short-range response controlled by the soliton size R_{sol}.
- The relative weight of the two terms determines the scale dependence of clustering and transport.

22 Effective Response Strength

The effective response strength is defined as:

$$G_{\text{eff}}(k, a) = -k^2 K(k, a). \tag{56}$$

This quantity measures how density perturbations source the response potential Φ at scale k. It is not a gravitational constant and does not represent any geometric structure.

22.1 Scale Dependence

For the minimal kernel:

$$G_{\text{eff}}(k, a) = -k^2 \left[\frac{A(a)}{k^2 + m^2(a)} + \frac{B(a)}{k^2 + \Lambda^2(a)} \right]. \tag{57}$$

Thus:

- At small k, the long-range term dominates.
- At large k, the short-range term dominates.
- The transition scale is set by $m(a)$ and $\Lambda(a)$.

23 Dual Projection and Slip

The continuum admits two distinct projections of the response kernel:

$$\Phi(k,a) = K(k,a)\,\rho(k,a), \tag{58}$$

$$\Psi(k,a) = K_2(k,a)\,\rho(k,a), \tag{59}$$

where the second projection is:

$$K_2(k,a) = K(k,a)\,\left[1+\epsilon(a)\,f(k,a)\right]. \tag{60}$$

The shape function is:

$$f(k,a) = \frac{k^2}{k^2+\Lambda^2(a)}. \tag{61}$$

The anisotropy parameter $\epsilon(a)$ determines the strength of the difference between the two projections.

23.1 Slip Parameter

The slip parameter is defined as:

$$\eta(k,a) = \frac{\Phi(k,a)}{\Psi(k,a)} = \frac{K(k,a)}{K_2(k,a)} = \frac{1}{1+\epsilon(a)\,f(k,a)}. \tag{62}$$

This quantity measures anisotropy in the internal response. It is not a geometric ratio and does not involve curvature.

23.2 Limiting Behaviour

- For $k \ll \Lambda(a)$: $f(k,a) \to 0$, so $\eta \to 1$.
- For $k \gg \Lambda(a)$: $f(k,a) \to 1$, so $\eta \to 1/(1+\epsilon)$.

Slip therefore increases toward small scales, reflecting sensitivity to finite soliton size.

24 Lensing Potential

The lensing potential is defined as:

$$\Phi_{\text{lens}}(k,a) = \frac{1}{2}\left[\Phi(k,a)+\Psi(k,a)\right]. \tag{63}$$

This projection governs wave-like propagation and is sensitive to the difference between the two kernel projections.

25 Scale-Dependent Response

The kernel structure produces natural scale dependence in clustering and transport:

- Long-range response dominates at small k.
- Short-range response dominates at large k.
- Slip introduces anisotropic suppression at high k.

These effects arise entirely from internal continuum structure. No geometric or force-based interpretation is required.

Part IV: Linear Perturbation Sector

26 Density Contrast

Define the density contrast as the fractional deviation of the inertial density from its homogeneous background value:

$$\delta(x,t) = \frac{\rho(x,t) - \rho_0(t)}{\rho_0(t)}. \tag{64}$$

In Fourier space:

$$\delta(k,t) = \frac{\rho(k,t)}{\rho_0(t)}. \tag{65}$$

The density contrast is not a fundamental field. It is a projection of the soliton ensemble and serves as a coarse-grained descriptor of inhomogeneity.

27 Coupling Perturbation

Define the coupling perturbation as:

$$\beta(x,t) = \frac{\alpha(x,t) - \alpha_0(t)}{\alpha_0(t)}. \tag{66}$$

In Fourier space:

$$\beta(k,t) = \frac{\alpha(k,t)}{\alpha_0(t)}. \tag{67}$$

The coupling perturbation measures deviations in the local response content of the soliton gas. It plays a central role in the feedback between density and response.

28 Velocity Divergence

Define the velocity divergence:

$$\theta(x,t) = \nabla \cdot u(x,t). \tag{68}$$

In Fourier space:

$$\theta(k,t) = ik \cdot u(k,t). \tag{69}$$

The velocity divergence describes the compressive or expansive behaviour of the soliton gas under coarse-grained transport.

29 Deterministic Growth Equation

The linearised continuity equation is:

$$\dot{\delta} + \frac{\theta}{a} = 0. \tag{70}$$

The linearised momentum equation is:

$$\dot{\theta} + H\theta = -\frac{k^2}{a}\psi - \frac{c_s^2 k^2}{a}\delta - \frac{\nu k^2}{a^2}\theta, \tag{71}$$

where ψ is the response potential perturbation.

The potential perturbation is:

$$\psi(k,a) = -G_{\text{eff}}(k,a)\,\rho_0(a)\,a^2\frac{\delta(k,a)}{k^2}. \tag{72}$$

Combining the continuity and momentum equations yields the deterministic growth equation:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\text{eff}}(k,a)\,\rho_0(a)\,[\delta + \chi\beta] - \frac{c_s^2 k^2}{a^2}\delta. \tag{73}$$

The parameter χ controls the strength of coupling feedback.

29.1 Interpretation

- The term $4\pi G_{\text{eff}}\rho_0\delta$ drives growth.
- The term $4\pi G_{\text{eff}}\rho_0\chi\beta$ introduces composition-dependent feedback.
- The term $c_s^2 k^2\delta/a^2$ suppresses growth at high k.

All terms arise from internal continuum structure. No geometric or force-based interpretation is required.

30 Scale-Dependent Growth

The kernel structure produces natural scale dependence in the growth of density perturbations. For the minimal kernel:

$$G_{\text{eff}}(k,a) = -k^2\left[\frac{A(a)}{k^2 + m^2(a)} + \frac{B(a)}{k^2 + \Lambda^2(a)}\right]. \tag{74}$$

Thus:

- At small k, long-range response dominates and growth is enhanced.
- At large k, short-range response dominates and growth is suppressed.
- The transition scale is set by $m(a)$ and $\Lambda(a)$.

30.1 Coupling Feedback

The coupling perturbation obeys:

$$\dot{\beta} + \left[\Gamma_\alpha(a) + D_\alpha(a)\frac{k^2}{a^2}\right]\beta = s(a)\,\delta. \tag{75}$$

The feedback term $\chi\beta$ in the growth equation introduces composition-dependent response:

- $\chi > 0$: feedback suppresses growth,
- $\chi < 0$: feedback enhances growth,
- $\chi = 0$: pure kernel-driven growth.

The effect is strongest at small k, where β is least damped.

31 Deterministic Transfer Function

Define the deterministic growth factor $D(k, a)$ by:

$$\delta(k, a) = D(k, a)\, \delta(k, a_{\text{init}}). \tag{76}$$

The deterministic transfer function is:

$$T_R(k, a) = \frac{D(k, a)}{D(k, a_{\text{init}})}. \tag{77}$$

The power spectrum evolves as:

$$P(k, a) = D^2(k, a)\, T_R^2(k, a)\, P_{\text{init}}(k). \tag{78}$$

The scale dependence of T_R is determined entirely by the kernel and the coupling feedback.

Part V: Stochastic Response Sector (Quantum Floor)

32 Origin of Stochasticity

Coarse-graining the soliton ensemble introduces irreducible fluctuations that cannot be removed by further smoothing. These fluctuations arise from:

1. finite soliton number per coarse cell,
2. variance in coupling content σ_α^2,
3. variance in inertial content ρ_0,
4. finite correlation length ξ_R.

These fluctuations are internal to the continuum. They do not represent external randomness, quantum fields, or metric fluctuations. They are the residual imprint of soliton discreteness on the coupling sector.

33 Stochastic Coupling Equation

The deterministic coupling perturbation equation becomes a stochastic differential equation:

$$\beta' + \left[\Gamma_\alpha(a) + D_\alpha(a)\frac{k^2}{a^2}\right]\beta = s(a)\,\delta + \Xi_\beta(k, a), \tag{79}$$

where Ξ_β is the unresolved remainder of the soliton ensemble after coarse-graining.

No new ontology is introduced. The stochastic term is a projection of existing internal variance.

34 Noise Statistics and Effective Fluctuation Scale

The noise has zero mean:

$$\langle \Xi_\beta(k,a) \rangle = 0, \tag{80}$$

and a two-point correlation:

$$\langle \Xi_\beta(k,a)\, \Xi_\beta(k',a') \rangle = N_\beta(a,k)\, \delta_D(\ln a - \ln a')\, \delta_D(k-k'). \tag{81}$$

The amplitude is:

$$N_\beta(a,k) = C_\beta\, \hbar_{\rm eff}(a)\, f_\beta(k), \tag{82}$$

where the effective fluctuation scale is:

$$\hbar_{\rm eff}(a) = \sigma_\alpha^2(a)\, \rho_0(a). \tag{83}$$

This quantity defines the *Quantum Floor*: the irreducible fluctuation level of the continuum.

35 Propagation of Stochasticity into δ

The density contrast inherits stochasticity through the $\chi\beta$ term in the growth equation:

$$\delta(k,a) = \delta_{\rm cl}(k,a) + \delta_{\rm noise}(k,a). \tag{84}$$

The power spectrum becomes:

$$P_\delta(k,a) = P_{\delta,\rm cl}(k,a) + P_{\delta,\rm noise}(k,a). \tag{85}$$

At large scales and early times:

$$P_{\delta,\rm cl} \gg P_{\delta,\rm noise}. \tag{86}$$

At small scales and late times:

$$P_{\delta,\rm cl} \to 0, \qquad P_{\delta,\rm noise} \to \text{finite constant}. \tag{87}$$

This constant is the noise floor of the continuum.

36 High-k Local Response Limit

For $k \gg 1/\xi_R$, the kernel becomes local:

$$K(k,a) \approx \frac{A(a)}{k^2}, \tag{88}$$

and the deterministic growth equation reduces to:

$$\delta'' + 2H\delta' + c_s^2 \frac{k^2}{a^2}\delta = 4\pi A(a)\, \rho_0(a)\, \chi\, \beta. \tag{89}$$

The deterministic part decays rapidly. The stochastic part drives a stationary solution.

37 Green's Function Solution

Define the operator:

$$L_\delta = \frac{d^2}{dt^2} + 2H\frac{d}{dt} + c_s^2\frac{k^2}{a^2}. \tag{90}$$

The stochastic solution is:

$$\delta_{\text{noise}}(k,t) = \int dt'\, G_\delta(t,t';k)\, S_\beta(t',k), \tag{91}$$

where:

$$S_\beta(t',k) = 4\pi A(a')\,\rho_0(a')\,\chi\,\Xi_\beta(k,t'). \tag{92}$$

For slowly varying $H(a)$ and large k:

$$G_\delta(t,t';k) \approx \left(\frac{a'}{a}\right)\exp\left[-\frac{c_s k}{a}(t-t')\right]. \tag{93}$$

38 Stationary Noise Spectrum

Using the noise statistics:

$$\langle \Xi_\beta(k,t)\,\Xi_\beta(k',t')\rangle = N_\beta(a,k)\,\delta_D(t-t')\,\delta_D(k-k'), \tag{94}$$

the noise-driven power spectrum becomes:

$$P_{\delta,\text{noise}}(k,a) \approx \frac{\left[4\pi A(a)\,\rho_0(a)\,\chi\right]^2 N_\beta(a,k)}{2c_s k/a}. \tag{95}$$

Substituting N_β and $\hbar_{\text{eff}}$:

$$P_{\delta,\text{noise}}(k,a) \approx \frac{\left[4\pi A(a)\,\rho_0(a)\,\chi\right]^2 C_\beta\,\sigma_\alpha^2(a)\,\rho_0(a)\,f_\beta(k)}{2c_s k/a}. \tag{96}$$

39 Physical Interpretation

The noise floor scales as:

$$P_{\delta,\text{noise}} \propto A(a)^2\,\rho_0(a)^3\,\sigma_\alpha^2(a)\,f_\beta(k)\,\frac{a}{k}. \tag{97}$$

This is a purely CCEF result:

- it depends only on internal continuum quantities,
- it arises from soliton discreteness,
- it contains no geometric or quantum-field-theoretic input,
- it defines the irreducible fluctuation level of δ at high k.

The continuum cannot be made perfectly smooth at small scales. Even when deterministic growth is suppressed, the soliton gas produces a finite residual variance. This is the CCEF-native analogue of a zero-point floor, arising entirely from internal variance and response structure.

Part VI: Merger Feedback and RG Flow

40 Merger-Driven Variance Evolution

The coupling variance $\sigma_\alpha^2(a)$ evolves under the combined influence of transport, mixing, and soliton mergers. In dense regions, mergers occur more frequently, reducing the diversity of soliton lineages and driving the coupling field toward universality.

The variance evolution equation is:

$$\frac{d\sigma_\alpha^2}{d\ln a} = -\Gamma_{\text{merger}}(\delta)\,\sigma_\alpha^2 + S_{\text{var}}(a), \tag{98}$$

where S_{var} represents any source of new variance (e.g., creation channels), and the merger-driven relaxation rate is:

$$\Gamma_{\text{merger}}(\delta) = \Gamma_0 \exp\left(\frac{\delta}{\delta_{\text{crit}}}\right). \tag{99}$$

40.1 Interpretation

- At low density contrast ($\delta \ll \delta_{\text{crit}}$), mergers are rare and σ_α^2 decays slowly.
- At high density contrast ($\delta \gtrsim \delta_{\text{crit}}$), mergers become efficient and σ_α^2 collapses rapidly.

This mechanism drives the system toward universality in dense regions while preserving non-universality in voids.

41 Density-Triggered RG Flow

The density contrast acts as the trigger for RG flow in the coupling variance. The RG equation for σ_α^2 is:

$$\frac{d\sigma_\alpha^2}{d\ln a} = -\Gamma_0 \exp\left(\frac{\delta}{\delta_{\text{crit}}}\right)\sigma_\alpha^2. \tag{100}$$

This flow has the following properties:

- It is not time-based; it is density-based.
- It introduces no tunable decay timescale.
- It is purely mechanistic, arising from soliton mergers.

41.1 Consequences

1. Variance decays rapidly in halos.
2. Variance remains high in voids.
3. The universe becomes spatially textured.

This mechanism is responsible for the Texture Transition.

42 Correlation Length Flow

The response correlation length $\xi_R(a)$ evolves under RG flow:

$$\frac{d\xi_R}{d\ln a} = \beta_\xi(\rho_0, \alpha_0, \sigma_\alpha^2). \tag{101}$$

A simple logistic form is:

$$\frac{d\xi_R}{d\ln a} = \beta_\xi\, \xi_R \left(1 - \frac{\xi_R}{\xi_{R,\max}}\right). \tag{102}$$

42.1 Interpretation

- At early times, ξ_R grows rapidly.
- Growth slows as ξ_R approaches its IR fixed point.
- The rate of internal reorganisation peaks at intermediate a.

This behaviour defines the internal time variable $\tau(a)$.

43 Soliton RG Flow

Soliton properties evolve under coarse-graining. The effective mass, size, and response content flow with the block scale b:

$$\frac{dM}{d\ln b} = \beta_M(M, R, q), \tag{103}$$

$$\frac{dR}{d\ln b} = \beta_R(M, R, q), \tag{104}$$

$$\frac{dq}{d\ln b} = \beta_q(M, R, q). \tag{105}$$

Stable soliton types correspond to fixed points of this flow.

43.1 Universal and Nonuniversal Spectra

- If the merger RG has a stable fixed point, the soliton population approaches a universal spectrum.
- If no stable fixed point exists, the soliton population retains memory of initial conditions, producing persistent tracer-dependent behaviour.

44 Fixed Points and Universality

A fixed point of the RG flow satisfies:

$$\beta_K = 0, \qquad \beta_\alpha = 0, \qquad \beta_A = 0, \qquad \beta_B = 0. \tag{106}$$

At a fixed point, the continuum exhibits scale-invariant response.

44.1 Types of Fixed Points

- **Long-range fixed point:**

$$K(r) \sim \frac{1}{r}. \tag{107}$$

- **Screened fixed point:**

$$K(r) \sim \frac{e^{-r/\xi_R}}{r}. \tag{108}$$

- **Mixed fixed point:** both long-range and short-range components survive.

45 Relevant and Irrelevant Directions

Perturbations around a fixed point evolve according to the eigenvalues of the linearised flow. Positive eigenvalues correspond to relevant directions; negative eigenvalues correspond to irrelevant directions.

This classification determines which features survive at large scales.

46 Feedback Loop Between Kernel and Soliton Sector

The response kernel and soliton spectrum interact through a feedback loop:

1. The kernel influences merger rates.
2. Mergers modify the coupling field.
3. The coupling field modifies the kernel amplitudes.

This loop determines the large-scale phase of the continuum.

47 Phase Boundaries

Phase transitions occur when:

- ξ_R diverges,
- σ_α^2 crosses a critical threshold,
- the kernel changes fixed-point class.

These transitions are internal reorganisations of the continuum.

Part VII: Phase Structure of the Continuum

48 Disordered Phase

At early times or in low-density regimes, the response correlation length ξ_R is short and the coupling variance σ_α^2 is large. The continuum behaves as a locally interacting medium with no large-scale coherence.

The defining characteristics are:

$$\xi_R \ll L_{\text{sys}}, \tag{109}$$

$$\sigma_\alpha^2 \text{ large}, \tag{110}$$

$$K(r) \sim \frac{e^{-r/\xi_R}}{r}. \tag{111}$$

48.1 Physical Behaviour

- Response is short-range and highly irregular.
- Density perturbations grow slowly and stochastically.
- The stochastic sector dominates small-scale behaviour.
- No geometry-like projection is possible.

This phase corresponds to the early universe and to void interiors at late times.

49 Percolation Phase

As the correlation length grows, the response begins to connect across larger regions. When ξ_R becomes comparable to the typical inter-soliton spacing, a percolation transition occurs.

The defining condition is:

$$\xi_R \approx \ell_{\text{sep}}. \tag{112}$$

49.1 Physical Behaviour

- Response pathways form extended networks.
- Transport becomes partially coherent.
- Density perturbations begin to grow more efficiently.
- Slip remains negligible.

This phase marks the onset of collective behaviour.

50 Texture Transition

The Texture Transition occurs when the merger-driven collapse of σ_α^2 becomes efficient in overdense regions. This transition is density-triggered and spatially inhomogeneous.

The defining condition is:

$$\Gamma_{\text{merger}}(\delta) \sim H, \tag{113}$$

i.e., mergers become competitive with bulk expansion.

50.1 Consequences

1. σ_α^2 collapses in halos.
2. σ_α^2 remains high in voids.
3. The universe becomes spatially textured.

This transition produces the characteristic void–halo asymmetry of CCEF.

51 Spatial Texture: Voids vs Halos

After the Texture Transition, the continuum exhibits strong environment-dependent variance:

$$\sigma_\alpha^2(\text{halo}) \ll \sigma_\alpha^2(\text{void}), \tag{114}$$

$$\xi_R(\text{halo}) \gg \xi_R(\text{void}). \tag{115}$$

51.1 Halo Behaviour

- Variance collapses due to mergers.
- Response becomes universal.
- Slip is suppressed.
- Growth is efficient.

51.2 Void Behaviour

- Variance remains high.
- Response is non-universal.
- Slip is enhanced.
- Growth is suppressed.

This spatial texture is a structural prediction of CCEF.

52 Coherent Phase

When the correlation length approaches the system size:

$$\xi_R \to L_{\text{sys}}, \tag{116}$$

the continuum enters a coherent phase.

52.1 Physical Behaviour

- The kernel becomes smooth.
- Transport becomes stable.
- Response distance becomes approximately quadratic.
- Geometry-like behaviour emerges as a projection.

This phase corresponds to the large-scale behaviour of the late-time universe.

53 Accelerating Bulk Phase

If the response pressure becomes sufficiently negative:

$$P_R < -\frac{\rho_0}{3}, \tag{117}$$

the bulk expansion accelerates.

53.1 Interpretation

- Acceleration is an internal relaxation effect.
- No dark energy or geometric curvature is required.
- The acceleration is driven by the response sector.

This phase corresponds to the observed late-time acceleration of the universe.

Part VIII: Observable Projections

54 Growth Rate

The growth rate is defined as:

$$f(a) = \frac{d \ln D(a)}{d \ln a}, \tag{118}$$

where $D(a)$ is the deterministic growth factor defined by:

$$\delta(k, a) = D(k, a)\, \delta(k, a_{\text{init}}). \tag{119}$$

In CCEF, the growth rate is scale dependent:

$$f(k, a) = \frac{d \ln D(k, a)}{d \ln a}. \tag{120}$$

54.1 Interpretation

- At small k, long-range response enhances growth.
- At large k, short-range response suppresses growth.
- The stochastic sector introduces a noise floor at high k.

This scale dependence is a structural prediction of CCEF.

55 Power Spectrum

The power spectrum is defined as:

$$P(k,a) = \langle |\delta(k,a)|^2 \rangle. \tag{121}$$

It decomposes into deterministic and stochastic components:

$$P(k,a) = P_{\rm cl}(k,a) + P_{\rm noise}(k,a). \tag{122}$$

55.1 Deterministic Component

$$P_{\rm cl}(k,a) = D^2(k,a)\, T_R^2(k,a)\, P_{\rm init}(k). \tag{123}$$

55.2 Stochastic Component

From the stochastic sector:

$$P_{\rm noise}(k,a) \approx \frac{[4\pi A(a)\,\rho_0(a)\,\chi]^2\, C_\beta\, \sigma_\alpha^2(a)\,\rho_0(a)\, f_\beta(k)}{2c_s k/a}. \tag{124}$$

55.3 Interpretation

- At small k, deterministic growth dominates.
- At large k, the noise floor dominates.
- The transition scale is determined by ξ_R and σ_α^2.

56 Slip Projection

The slip parameter is:

$$\eta(k,a) = \frac{\Phi(k,a)}{\Psi(k,a)} = \frac{1}{1+\epsilon(a)\, f(k,a)}. \tag{125}$$

56.1 Interpretation

- Slip is negligible at small k.
- Slip increases toward large k.
- Slip is enhanced in voids (high σ_α^2).
- Slip is suppressed in halos (low σ_α^2).

This behaviour is a direct consequence of finite soliton size and anisotropic response.

57 Lensing vs Clustering

The lensing potential is:

$$\Phi_{\text{lens}}(k,a) = \frac{1}{2}\left[\Phi(k,a) + \Psi(k,a)\right]. \tag{126}$$

Because Ψ includes anisotropic suppression:

$$\Phi_{\text{lens}} < \Phi \quad \text{for} \quad k \gg \Lambda(a). \tag{127}$$

57.1 Consequences

- Lensing underestimates clustering at high k.
- The discrepancy grows as $\epsilon(a)$ increases.
- This produces an apparent suppression of S_8.

This mechanism resolves the weak-lensing tension without modifying matter content.

58 Equivalence-Residuals

If different soliton types have different coupling strengths, they experience different effective accelerations:

$$a_{\text{eff}} \propto \alpha. \tag{128}$$

This produces tracer-dependent clustering:

$$\delta_{\text{tracer}}(k,a) = \delta(k,a)\ \left[1 + \Delta_\alpha(k,a)\right]. \tag{129}$$

58.1 Interpretation

- Not a violation of equivalence.
- A projection of internal composition.
- Controlled by σ_α^2 and merger history.

59 Internal-Time Redshift

In CCEF, observed redshift is a composite of:

$$1 + z_{\text{obs}} = (1 + z_{\text{exp}})(1 + z_\tau), \tag{130}$$

where:

$$1 + z_{\text{exp}} = \frac{1}{a}, \tag{131}$$

and the internal-time contribution is:

$$1 + z_\tau(a) = 1 + C_\tau\left[\tau(a) - \tau_{\text{ref}}\right]. \tag{132}$$

59.1 Internal Time

The internal time variable is:

$$\tau(a) = \frac{\xi_R(a) - \xi_{R,\text{init}}}{\xi_{R,\max} - \xi_{R,\text{init}}}. \tag{133}$$

Its rate of change is:

$$\frac{d\tau}{d\ln a} = \frac{1}{\xi_{R,\max} - \xi_{R,\text{init}}} \frac{d\xi_R}{d\ln a}. \tag{134}$$

59.2 Interpretation

- At early times, $d\tau/d\ln a$ is large.
- Internal evolution outpaces expansion.
- High-redshift galaxies are internally older than their expansion age.

This resolves the JWST high-redshift mass and maturity anomalies.

Appendix A: CCEF-Lite v1.0 Specification

A Ontological Core

The CCEF-Lite v1.0 ontology is defined by a single constrained continuum field $n(x,t)$ satisfying:

$$|n(x,t)| = 1. \tag{135}$$

No geometric, metric, or spacetime structure is assumed. The continuum is the only ontological entity. All physical structure arises from internal excitations, deformations, and topological configurations of $n(x,t)$.

The vacuum configuration is:

$$n(x,t) = n_0. \tag{136}$$

B Constraint and Configuration Manifold

The unit-norm constraint restricts the field to a nonlinear manifold. All admissible variations must preserve:

$$n \cdot n = 1. \tag{137}$$

This constraint is intrinsic and defines the allowed internal degrees of freedom.

C Internal Evolution Principle

All evolution is internal to the continuum. There is no external time, no external law of motion, and no background geometry. The continuum redistributes its internal configuration according to its variational structure.

Quantities such as energy, momentum, and pressure are emergent bookkeeping constructs.

D Variational Structure

The action is:

$$S = \int d^4x \left[\frac{1}{2} \partial_\mu n \cdot \partial^\mu n - V(n) - \lambda(x,t) \left(|n|^2 - 1 \right) \right], \tag{138}$$

where λ enforces the constraint.

Variation yields:

$$\partial_\mu \partial^\mu n + \frac{\partial V}{\partial n} + \lambda n = 0. \tag{139}$$

E Linear Excitation Sector

Small deviations around the vacuum are:

$$n = n_0 + \pi, \tag{140}$$

with:

$$n_0 \cdot \pi = 0. \tag{141}$$

Linearised dynamics give:

$$\Box \pi = 0. \tag{142}$$

These are tangent-space wave modes of the constrained continuum.

F Higher-Order Structure

Beyond the linear regime, additional internal operators appear in the effective description, including:

- quartic derivative terms,
- mixed contractions,
- nonlinear constraint corrections.

These modify dispersion and stability but remain internal to the continuum.

G Response Kernel

The internal response is encoded in a kernel $K(k,a)$. The minimal family is:

$$K(k,a) = \frac{A(a)}{k^2 + m^2(a)} + \frac{B(a)}{k^2 + \Lambda^2(a)}, \tag{143}$$

with:

$$m(a) = \frac{1}{\xi_R(a)}, \tag{144}$$

$$\Lambda(a) = \frac{1}{R_{\text{sol}}(a)}, \tag{145}$$

$$A(a) = C_\alpha \alpha_0^2(a) \rho_0(a), \tag{146}$$

$$B(a) = C_\sigma \sigma_\alpha^2(a) \rho_0(a). \tag{147}$$

H Topological Sector

The continuum supports stable topological configurations with:

$$\pi_3(S^2) = \mathbb{Z}. \tag{148}$$

Topological charge Q labels soliton species. Matter is not a separate ontology; it is persistent topology of $n(x,t)$.

I Soliton Stability

Solitons satisfy Derrick-type stability:

$$E(\lambda) = \lambda E_2 + \lambda^{-1} E_4. \tag{149}$$

The stable size is:

$$\lambda^* = \sqrt{\frac{E_4}{E_2}}. \tag{150}$$

J Emergent Inertia

A soliton with centre $X(t)$ behaves as:

$$T = \frac{1}{2} M \dot{X}^2, \tag{151}$$

where M is the soliton energy.

Inertia is emergent from internal deformation energy.

K Long-Range Response

In dilute regimes:

$$\nabla^2 \pi = 0, \tag{152}$$

with solutions:

$$\pi(r) = \frac{B}{r}. \tag{153}$$

This produces inverse-distance response behaviour.

L Coarse-Grained Fields

Coarse-graining over a smoothing volume ΔV yields:

$$\rho(x,t) = \frac{1}{\Delta V} \sum_i M_i W(x - X_i), \tag{154}$$

$$\sigma(x,t) = \frac{1}{\Delta V} \sum_i Q_i W(x - X_i), \tag{155}$$

$$u(x,t) = \frac{\sum_i P_i W(x - X_i)}{\sum_i M_i W(x - X_i)}. \tag{156}$$

These fields are projections of the soliton ensemble.

M Coupling Field

The effective coupling is:

$$\alpha(x,t) = \frac{\sigma(x,t)}{\rho(x,t)}. \tag{157}$$

If α is uniform, response is universal. If α varies, response is composition dependent.

N Transport Equations

Inertial density satisfies:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0. \tag{158}$$

Momentum balance is:

$$\rho(\partial_t u + u \cdot \nabla u) = -\nabla P - \rho \nabla \Phi + \nu \nabla^2 u + F_{\text{corr}}. \tag{159}$$

O Response Potential

The potential is:

$$\Phi(x,t) = \int K(x - x')\rho(x',t)\, d^3x'. \tag{160}$$

If $K(r) \sim 1/r$, then:

$$\nabla^2 \Phi = -4\pi C \rho. \tag{161}$$

P Background Evolution

In a homogeneous regime:

$$\rho = \rho_0(t), \quad \sigma = \sigma_0(t), \quad \alpha = \alpha_0(t). \tag{162}$$

With $u = Hx$:

$$\dot{\rho}_0 + 3H\rho_0 = 0, \tag{163}$$

so:

$$\rho_0 \propto a^{-3}. \tag{164}$$

Q Coupling Evolution

The mean coupling evolves as:

$$\dot{\alpha}_0 = -\Gamma_\alpha(\rho_0)\, \alpha_0 + S_0(\rho_0). \tag{165}$$

R Bulk Expansion Closure

The expansion rate satisfies:

$$H^2 = F(\rho_0, \alpha_0, P_R, \Sigma). \tag{166}$$

Acceleration occurs when:

$$P_R < -\frac{\rho_0}{3}. \tag{167}$$

S Linear Perturbations

Define:

$$\delta = \frac{\rho - \rho_0}{\rho_0}, \qquad \beta = \frac{\alpha - \alpha_0}{\alpha_0}, \qquad \theta = \nabla \cdot u. \tag{168}$$

Continuity:

$$\dot{\delta} + \frac{\theta}{a} = 0. \tag{169}$$

Momentum:

$$\dot{\theta} + H\theta = -\frac{k^2}{a}\psi - \frac{c_s^2 k^2}{a}\delta - \frac{\nu k^2}{a^2}\theta. \tag{170}$$

Potential:

$$\psi(k) = -G_{\text{eff}}(k, a)\, \rho_0 a^2 \frac{\delta}{k^2}. \tag{171}$$

Coupling perturbation:

$$\dot{\beta} + \left[\Gamma_\alpha + D_\alpha \frac{k^2}{a^2}\right] \beta = s(a)\, \delta. \tag{172}$$

Growth equation:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\text{eff}}(k, a)\, \rho_0(\delta + \chi\beta) - \frac{c_s^2 k^2}{a^2}\delta. \tag{173}$$

T Projection Layer

Observable quantities include:

- density contrast δ,
- velocity divergence θ,
- response potentials Φ and Ψ,
- lensing potential Φ_{lens},
- growth rate f,
- power spectrum $P(k)$,
- halo profiles,
- slip parameter η.

These are projections of the underlying continuum, not fundamental fields.

Appendix B: Stochastic Derivations

U Stochastic Origin of the Coupling Perturbation

The coupling perturbation $\beta(k, a)$ receives contributions from unresolved soliton discreteness. Coarse-graining over a finite volume ΔV leaves a residual fluctuation term Ξ_β, defined by:

$$\Xi_\beta(k, a) = \frac{1}{\alpha_0(a)} \frac{1}{\rho_0(a)} \sum_{i \in \Delta V} [Q_i - \alpha_0(a) M_i] \, e^{-ik \cdot X_i}. \tag{174}$$

This term has zero mean:

$$\langle \Xi_\beta(k, a) \rangle = 0, \tag{175}$$

and nonzero variance due to finite soliton number and finite coupling variance σ_α^2.

V Stochastic Coupling Equation

The deterministic coupling equation is:

$$\beta' + \left[\Gamma_\alpha(a) + D_\alpha(a) \frac{k^2}{a^2} \right] \beta = s(a) \, \delta. \tag{176}$$

Including the unresolved fluctuation term yields:

$$\beta' + \left[\Gamma_\alpha(a) + D_\alpha(a) \frac{k^2}{a^2} \right] \beta = s(a) \, \delta + \Xi_\beta(k, a). \tag{177}$$

This is a linear stochastic differential equation.

W Noise Correlation Structure

The noise is assumed Markovian in $\ln a$, with correlation:

$$\langle \Xi_\beta(k, a) \, \Xi_\beta(k', a') \rangle = N_\beta(a, k) \, \delta_D(\ln a - \ln a') \, \delta_D(k - k'). \tag{178}$$

The amplitude is:

$$N_\beta(a, k) = C_\beta \, \hbar_{\text{eff}}(a) \, f_\beta(k), \tag{179}$$

where the effective fluctuation scale is:

$$\hbar_{\text{eff}}(a) = \sigma_\alpha^2(a) \, \rho_0(a). \tag{180}$$

This quantity defines the *Quantum Floor* of the continuum.

X Solution of the Stochastic β Equation

The homogeneous solution is:

$$\beta_{\text{hom}}(a) = \beta_{\text{init}} \exp\left[- \int_{a_{\text{init}}}^{a} \left(\Gamma_\alpha(a') + D_\alpha(a') \frac{k^2}{a'^2} \right) d \ln a' \right]. \tag{181}$$

The particular solution is:

$$\beta_{\rm part}(k,a) = \int_{a_{\rm init}}^{a} G_\beta(a,a';k)\,[s(a')\,\delta(k,a') + \Xi_\beta(k,a')]\,d\ln a', \tag{182}$$

where the Green's function is:

$$G_\beta(a,a';k) = \exp\left[-\int_{a'}^{a}\left(\Gamma_\alpha(a'') + D_\alpha(a'')\frac{k^2}{a''^2}\right)d\ln a''\right]. \tag{183}$$

Y Propagation into the Density Contrast

The growth equation is:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\rm eff}(k,a)\,\rho_0(a)\,[\delta + \chi\beta] - \frac{c_s^2 k^2}{a^2}\delta. \tag{184}$$

Decompose:

$$\delta = \delta_{\rm cl} + \delta_{\rm noise}. \tag{185}$$

The noise-driven part satisfies:

$$L_\delta\,\delta_{\rm noise} = 4\pi G_{\rm eff}(k,a)\,\rho_0(a)\,\chi\,\beta_{\rm noise}, \tag{186}$$

where:

$$L_\delta = \frac{d^2}{dt^2} + 2H\frac{d}{dt} + c_s^2\frac{k^2}{a^2}. \tag{187}$$

Z Green's Function for δ

The Green's function satisfies:

$$L_\delta\,G_\delta(t,t';k) = \delta_D(t-t'). \tag{188}$$

For slowly varying $H(a)$ and large k:

$$G_\delta(t,t';k) \approx \left(\frac{a'}{a}\right)\exp\left[-\frac{c_s k}{a}(t-t')\right]. \tag{189}$$

Thus:

$$\delta_{\rm noise}(k,t) = \int dt'\,G_\delta(t,t';k)\,4\pi G_{\rm eff}(k,a')\,\rho_0(a')\,\chi\,\beta_{\rm noise}(k,t'). \tag{190}$$

Noise-Driven Power Spectrum

Using:

$$\langle \Xi_\beta(k,t)\,\Xi_\beta(k',t')\rangle = N_\beta(a,k)\,\delta_D(t-t')\,\delta_D(k-k'), \tag{191}$$

and substituting the Green's function, the noise-driven power spectrum is:

$$P_{\delta,\rm noise}(k,a) = \int dt'\,\left[4\pi G_{\rm eff}(k,a')\,\rho_0(a')\,\chi\right]^2 G_\delta^2(t,t';k)\,N_\beta(a',k). \tag{192}$$

For large k, the integral is dominated by the exponential decay of G_δ, yielding:

$$P_{\delta,\rm noise}(k,a) \approx \frac{\left[4\pi A(a)\,\rho_0(a)\,\chi\right]^2 N_\beta(a,k)}{2c_s k/a}. \tag{193}$$

Stationary Noise Floor

Substituting $N_\beta(a,k)$ and $\hbar_{\rm eff}(a)$:

$$P_{\delta,\rm noise}(k,a) \approx \frac{[4\pi A(a)\,\rho_0(a)\,\chi]^2\, C_\beta\, \sigma_\alpha^2(a)\,\rho_0(a)\, f_\beta(k)}{2c_s k/a}. \tag{194}$$

This is the *stationary noise spectrum* of the continuum.

.1 Interpretation

- The noise floor is unavoidable.
- It arises from soliton discreteness and coupling variance.
- It dominates small-scale behaviour at late times.
- It is a structural prediction of CCEF.

Appendix C: Internal-Time Redshift Derivations

Correlation-Length Flow and Internal Time

The response correlation length $\xi_R(a)$ evolves under RG flow:

$$\frac{d\xi_R}{d\ln a} = \beta_\xi(\rho_0, \alpha_0, \sigma_\alpha^2). \tag{195}$$

A convenient phenomenological form is the logistic flow:

$$\frac{d\xi_R}{d\ln a} = \beta_\xi\, \xi_R \left(1 - \frac{\xi_R}{\xi_{R,\rm max}}\right), \tag{196}$$

where $\xi_{R,\rm max}$ is the IR fixed point of the response sector.

.1 Solution of the Logistic Flow

The solution is:

$$\xi_R(a) = \frac{\xi_{R,\rm max}}{1 + \left(\frac{\xi_{R,\rm max}}{\xi_{R,\rm init}} - 1\right) a^{-\beta_\xi}}. \tag{197}$$

At early times ($a \ll 1$):

$$\xi_R(a) \approx \xi_{R,\rm init}\, a^{\beta_\xi}. \tag{198}$$

At late times ($a \to 1$):

$$\xi_R(a) \to \xi_{R,\rm max}. \tag{199}$$

Definition of Internal Time

Internal time $\tau(a)$ is defined as the normalised progression of the correlation length:

$$\tau(a) = \frac{\xi_R(a) - \xi_{R,\rm init}}{\xi_{R,\rm max} - \xi_{R,\rm init}}. \tag{200}$$

Thus:

$$\tau(a_{\rm init}) = 0, \qquad \tau(a=1) = 1. \tag{201}$$

.1 Interpretation

- τ measures the internal reorganisation of the continuum.
- It is not geometric time.
- It is not expansion time.
- It is the cumulative change in the response sector.

Internal-Time Dilatation

Different epochs evolve internally at different rates. The rate of internal evolution relative to expansion is:

$$\frac{d\tau}{d\ln a} = \frac{1}{\xi_{R,\max} - \xi_{R,\text{init}}} \frac{d\xi_R}{d\ln a}. \tag{202}$$

Substituting the logistic flow:

$$\frac{d\tau}{d\ln a} = \frac{\beta_\xi\, \xi_R(a) \left(1 - \frac{\xi_R(a)}{\xi_{R,\max}}\right)}{\xi_{R,\max} - \xi_{R,\text{init}}}. \tag{203}$$

.1 Behaviour

- At early times, $d\tau/d\ln a$ is large.
- Internal evolution outpaces expansion.
- At late times, $d\tau/d\ln a \to 0$.

This produces an internal-time dilation effect.

Composite Redshift Formula

Observed redshift is a composite of expansion and internal-time contributions:

$$1 + z_{\text{obs}} = (1 + z_{\text{exp}})(1 + z_\tau). \tag{204}$$

The expansion contribution is:

$$1 + z_{\text{exp}} = \frac{1}{a}. \tag{205}$$

The internal-time contribution is:

$$1 + z_\tau(a) = 1 + C_\tau \left[\tau(a) - \tau_{\text{ref}}\right], \tag{206}$$

where C_τ is a dimensionless projection coefficient.

.1 Interpretation

- Internal evolution modifies observed redshift.
- High-redshift galaxies can be internally older than their expansion age.
- This resolves the JWST high-redshift maturity anomalies.

Effective Lookback Time

Define the effective lookback time:

$$t_{\text{eff}}(z) = \int_{a(z)}^{1} \frac{d\ln a'}{H(a')} \left[1 + \frac{d\tau}{d\ln a'}\right]. \tag{207}$$

The second term increases the effective age of high-redshift objects.

Summary of Internal-Time Effects

1. $\xi_R(a)$ grows rapidly at early times.
2. $\tau(a)$ increases faster than $\ln a$.
3. Internal evolution produces an additional redshift component.
4. High-redshift galaxies appear older than expected.
5. No exotic physics or modified expansion is required.

Appendix D: Toy Models and Numerical Demonstrations

Overview

This appendix presents a set of simplified toy models illustrating the behaviour of the CCEF continuum under deterministic growth, coupling feedback, stochastic forcing, and internal-time evolution. These models are not approximations of the full theory; they are pedagogical demonstrations of specific mechanisms.

All models assume:

- homogeneous background density $\rho_0(a)$,
- minimal kernel family $K(k, a)$,
- linear perturbation regime,
- no geometric interpretation of a or k.

Toy Model I: Deterministic Growth

Consider the deterministic growth equation:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\text{eff}}(k, a)\, \rho_0(a)\, \delta - \frac{c_s^2 k^2}{a^2}\delta. \tag{208}$$

Assume:

$$H(a) = H_0 a^{-3/2}, \tag{209}$$

$$\rho_0(a) = \rho_{0,\text{init}} a^{-3}, \tag{210}$$

$$G_{\text{eff}}(k, a) = \frac{G_0}{1 + (k/k_0)^2}. \tag{211}$$

.1 Solution

The growth factor satisfies:

$$D'' + \frac{3}{2a}D' = \frac{3}{2a^2}\frac{1}{1+(k/k_0)^2}D. \tag{212}$$

For $k \ll k_0$:

$$D(a) \propto a. \tag{213}$$

For $k \gg k_0$:

$$D(a) \propto a^{1/(1+(k/k_0)^2)}. \tag{214}$$

This illustrates scale-dependent growth.

Toy Model II: Coupling Feedback Sweep

The coupling perturbation obeys:

$$\beta' + \Gamma_\alpha \beta = s\,\delta. \tag{215}$$

The growth equation becomes:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G_{\text{eff}}\rho_0(\delta + \chi\beta). \tag{216}$$

.1 Feedback Parameter Sweep

Let:

$$\chi \in [-1, 1]. \tag{217}$$

- $\chi = -1$: maximal enhancement.
- $\chi = 0$: pure kernel-driven growth.
- $\chi = +1$: maximal suppression.

.2 Example Solutions

For constant H, G_{eff}, and Γ_α:

$$\delta(a) = C_1 a^{p_+} + C_2 a^{p_-}, \tag{218}$$

$$p_\pm = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{3}{2}(1 + \chi s/\Gamma_\alpha)}. \tag{219}$$

This demonstrates how coupling feedback modifies growth exponents.

Toy Model III: Stochastic Noise Floor

The stochastic β-equation is:

$$\beta' + \Gamma_\alpha \beta = \Xi_\beta, \tag{220}$$

with:

$$\langle \Xi_\beta(a)\,\Xi_\beta(a')\rangle = N_\beta\,\delta_D(\ln a - \ln a'). \tag{221}$$

.1 Solution

$$\beta(a) = \int_{a_{\rm init}}^{a} e^{-\Gamma_\alpha(\ln a - \ln a')} \Xi_\beta(a')\, d\ln a'. \tag{222}$$

The variance is:

$$\langle \beta^2(a) \rangle = \frac{N_\beta}{2\Gamma_\alpha}. \tag{223}$$

.2 Propagation into δ

For large k:

$$\delta_{\rm noise}(a) \approx \frac{4\pi A \rho_0 \chi}{c_s k/a} \beta(a). \tag{224}$$

Thus:

$$P_{\delta,\rm noise}(k) \propto \frac{1}{k}. \tag{225}$$

This reproduces the stationary noise floor.

Toy Model IV: Internal-Time Simulation

Assume logistic correlation-length flow:

$$\frac{d\xi_R}{d\ln a} = \beta_\xi \xi_R \left(1 - \frac{\xi_R}{\xi_{R,\rm max}}\right). \tag{226}$$

.1 Internal Time

$$\tau(a) = \frac{\xi_R(a) - \xi_{R,\rm init}}{\xi_{R,\rm max} - \xi_{R,\rm init}}. \tag{227}$$

.2 Effective Redshift

$$1 + z_{\rm obs} = \frac{1}{a}\left[1 + C_\tau(\tau - \tau_{\rm ref})\right]. \tag{228}$$

This model reproduces the internal-time redshift shift.

Example Numerical Code

The following pseudocode illustrates numerical integration of the stochastic growth equation:

```
# Parameters
Gamma = 0.5
chi = -0.3
cs = 0.01
A = 1.0
rho0 = 1.0
Nbeta = 0.02
```

```
# Initialize
delta = 1e-5
beta = 0.0

for step in range(Nsteps):
    Xi = sqrt(Nbeta) * random.gauss(0,1)
    beta += -Gamma * beta * dloga + Xi * sqrt(dloga)
    delta += (4*pi*A*rho0*(delta + chi*beta) - cs*k*k*delta) * dloga
```

This demonstrates the emergence of the noise floor.

Summary

1. Deterministic growth is scale dependent.
2. Coupling feedback modifies growth exponents.
3. Stochastic forcing produces a stationary noise floor.
4. Internal-time evolution modifies observed redshift.

These toy models illustrate the core mechanisms of CCEF in simplified settings.